

Make This Simple Graphical LCD SCOPE

FAYAZ HASSAN

This scope reads input signal and displays the output waveform on a graphical liquid crystal display (GLCD) having 128×64 -pixel resolution. The signal is fed to ADC pin of ATmega16 microcontroller (MCU) through an operational amplifier (LM358). The ADC continuously reads the input voltage with preset delay and displays the same on GLCD. This circuit can measure the input frequency signal up to 250kHz.

Circuit and working

Circuit diagram of the simple GLCD scope is shown in Fig. 1.

GLCD. The GLCD (JHD12864E) has $0.48\text{mm} \times 0.48\text{mm}$ pixels size and approximately $6.6\text{cm} \times 3.3\text{cm}$ screen area. The screen is logically divided into two panes of 64×64 pixels on the left and 64×64 pixels on the right; this makes 128×64 pixels in total. The left-side pane is selected using CS1 pin, whereas the right-side pane is selected using CS2 pin. Data and command to GLCD are transmitted in 8-bit format by selecting RS pin settings as high and low, respectively. Contrast of the display is controlled at pin 3 of GLCD using preset VR1 connected between pin 18 (-10V DC) and ground (GND) of GLCD.

Sampling. ATmega16 is the heart of the circuit and signal voltage is read at its pin 40 (ADC channel 0 or PA0) based on the delay set as sampling time by potmeters VR2 and VR3. Sampling time and sampling rate are displayed on top of the GLCD as delay between two samples in microseconds (μs) and samples per seconds (S/S). Frequency in Hz for the total screen (for 128 samples) is displayed at the bottom-left corner. Reference ADC voltage is selected as internal 2.56V and ADC clock is set as 250kHz. ADC converted value is of 8-bit accuracy (reads 0 to 255). The MCU reads 256 samples and displays selected 128 samples on the GLCD

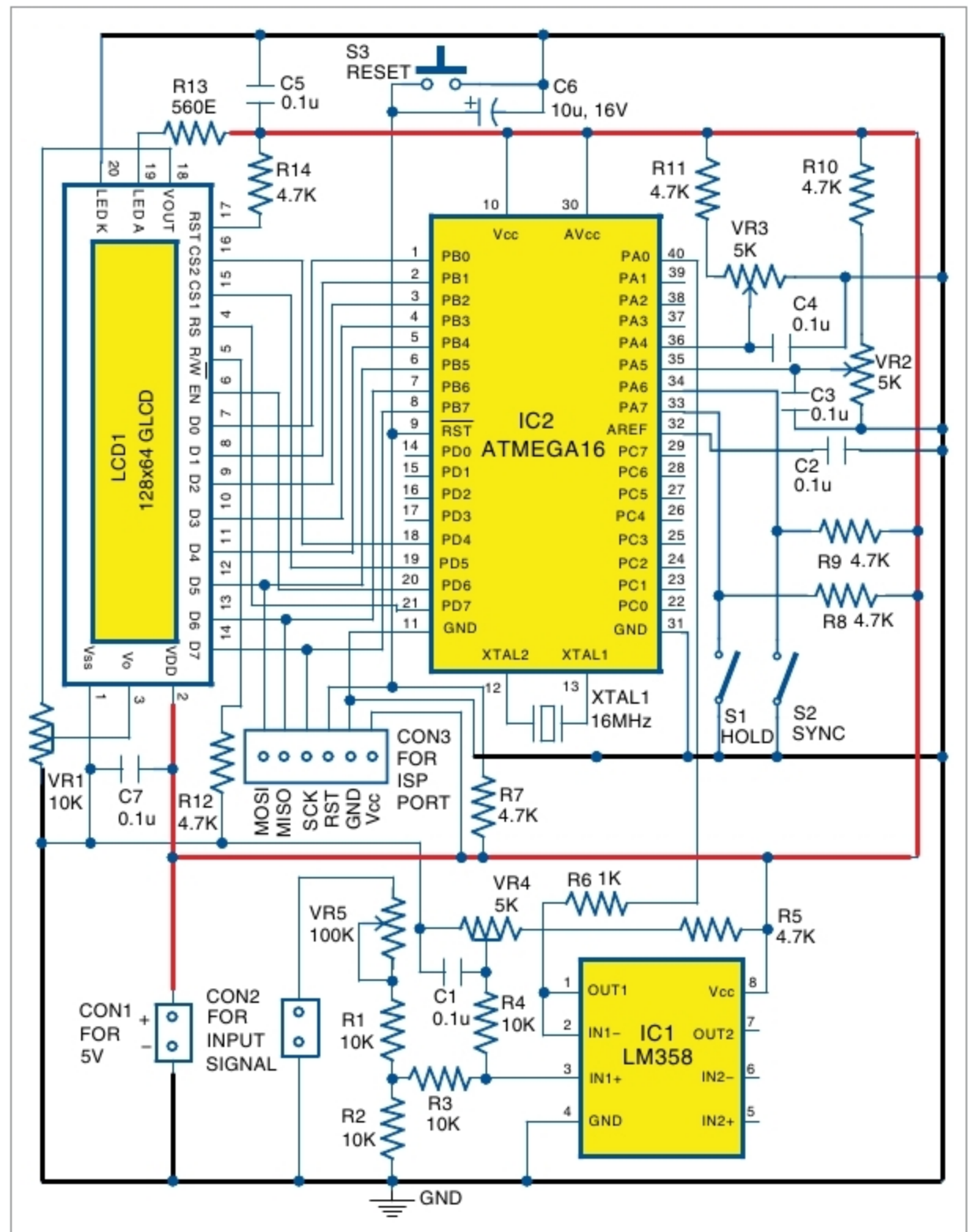


Fig. 1: Circuit diagram of the GLCD scope

depending on the status of sync switch S2.

S2 is used for synchronising the signal with the display. If S2 is closed (logic low), then the display starts from the maximum value of 128 samples. On the other hand, if S2 is open (logic high), the program makes the MCU display the first 128 samples.

If hold switch S1 is closed (logic low), the program stops reading the

signal value and the last sample data is continuously displayed. To continue sampling, open S1 (logic high).

LM358 is configured as the summing amplifier with unity gain. One of the inputs is set to have mid value of the maximum measurable voltage, that is, 1.28V, which can be adjusted using preset VR4. To adjust input voltage level, use VR5. Maximum input voltage is $\pm 25\text{V}$ and can be adjusted